Native fork join: work stealing by context switching

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Abstract

TaskPaRTS is a C++ library for fork-join parallelism that requires no compiler support. At its core is a technique we call *native fork join*, which delivers work-stealing scheduling by context switching. A fork saves the continuation of the parent task as a handful of machine registers that point into the parent's own stack. When the join finds that the second branch was stolen, the worker neither blocks nor runs other work on top of the parent's frame: it leaves the parent's stack suspended and returns to its scheduler. Whichever worker finishes the last branch then resumes the parent, possibly on a different thread. When nothing is stolen, both branches run on the parent's stack, like ordinary calls, and the fork allocates no memory. In this short paper, we present the algorithm in pseudocode, explain why it is safe, and describe what it means for programs that use it. A companion implementation, micro taskparts, implements the same algorithm in a few hundred lines of C++.

1 Challenge: the join, in a library

We are to implement binary fork join, which runs two thunks, possibly in parallel, and returns once both have finished.

```
fork2join(f1, f2 : () → void) → void
```

For example:

```
fib(n : int) → int {
  if (n < 2) return n
  fork2join(λ () ⇒ {a = fib(n-1)},
            λ () ⇒ {b = fib(n-2)})
  return a + b }
```

Work stealing [5, 8] is well suited to scheduling such programs. Our constraint is that we are writing a library: there is no compiler support, and client code is ordinary C++, with lambdas for thunks.

A library work stealer typically implements `fork2join` by *child stealing*: the second branch becomes a task that thieves may take.

```
fork2join(f1, f2 : () → void) → void {
  t = new_task(f2); push(t)   // t is now stealable
  f1()
  if (pop() == t) { f2(); return }   // not stolen
  while (!t.done)                    // stolen: wait for thief
    { u = steal(); if (u != null) u.run() } }
```

The common case, where `t` is not stolen, is cheap. The difficulty is the join when `t` was stolen. The parent's frame sits on the worker's thread stack, and must stay there until the join completes. So the worker can either block, and sit idle, or, as TBB and ParlayLib do [4, 9], *help*: run other tasks on top of the blocked frame, as in the loop above. Helping keeps the worker busy, but the parent can resume only once the worker has unwound back down to its frame, even if the thief finished long before. Cilk [7, 14] avoids waiting by stealing the parent's continuation instead of the child. But then the thief must resume a frame that sits on the victim's stack, which in Cilk relies on compiler support and a cactus stack [10].

Native fork join, which is at the core of TaskPaRTS [12], takes a different route. When a worker finds at a join that the second branch was stolen, it *suspends the parent's stack* and returns to its scheduler, which looks for other work. The parent's continuation is a saved register context that points into the suspended stack. Whichever worker finishes the last branch restores this context, and so the parent continues on that worker's thread. No worker waits at a join, and no two workers ever run on the same stack at once.

We present the algorithm in Section 2, trace its two cases in Section 3, explain why it is safe in Section 4, and discuss its costs and its consequences for client code in Section 5. The appendices give machine-level details, the work-stealing deque, a map from the pseudocode to TaskPaRTS and to micro taskparts (a minimal C++ implementation that accompanies this paper), and notes on performance. The last appendix compares native fork join with other ways of managing stacks under work stealing.

2 Native fork join

Contexts. Our one low-level primitive is the *context*. A context holds what the C calling convention requires a function call to preserve: the callee-saved registers, the stack pointer, and a return address. That is 8 words on x86-64 and 21 on AArch64 (Appendix A). `save(`k`)` stores its caller's context in k and returns **`false`**; `restore(`k`)` resumes that call to `save`, which this time returns **`true`**. The pair behaves like `setjmp` and `longjmp`, except that the thread that restores a context need not be the one that saved it, and the context may point into any stack. To the compiler, `save` is an ordinary call (one that returns twice, like `setjmp`), so it spills any live caller-saved registers before the call. As a result, a save or a restore is a handful of loads and stores, with no system call. We write `swap(`k_1`, `k_2`)` for **`if`** `(!save(`k_1`)) restore(`k_2`)`,

which saves the current context in k_1 and switches to k_2. Micro taskparts does both in a single routine, which on Apple silicon makes a fork more than twice as fast (Appendix D).

Vertices and workers. Following the DAG calculus [1], the scheduler represents a computation as a DAG whose vertices each have an *in-counter*, which counts unfinished predecessors, and an *out-set*, which, in fork join, is a single successor (Figure 1). A vertex also has a context, `k`, and an action, `act`, which tells the scheduler what to do with the vertex. `Init` means that the vertex has never run; `Resume`, that it is suspended inside `fork2join`, with its context saved in `k`; and `Finish`, that it will have finished the next time it hands control back to the scheduler.

Each worker runs a scheduler loop on its thread's own stack. While the worker runs a vertex, `cur`, it keeps the loop's context in `k`. Client code runs on separate *task stacks*. We write `me` for the worker that runs the caller, which we look up again at each use, because the worker can change across a call to `fork2join` (Section 5). Each worker also holds its ready vertices: `schedule(`v`)` makes v ready, `pop()` takes back the vertex scheduled most recently, if any, and `steal()` takes the oldest ready vertex of another worker, chosen at random. The vertex scheduled most recently can never be stolen (Appendix B), and `fork2join` relies on this.

The scheduler. The scheduler loop (Figure 1) is the only place where vertices start, resume, and finish. It pops a vertex, or else steals one and makes it ready, and then makes the vertex current. If the vertex has never run, the loop gives it a fresh stack and a context that, once restored, calls `enter` at the top of that stack. `enter` runs the body of the vertex, and then switches to the scheduler of whichever worker it is on by then. The loop then switches to the vertex, and regains control when the vertex yields. If the action of the vertex is then `Finish`, the loop notifies its successor: it decrements the successor's in-counter, and the worker whose decrement reaches zero schedules the successor. A call to `fork2join` from outside the scheduler starts the workers with a root vertex that makes the call, and the computation ends when the root, which has no successor, finishes.

Fork join. In `fork2join` (Figure 2), the parent, v_0, is the current vertex. The function creates two vertices, v_1 and v_2, both with v_0 as their successor. v_2 runs f_2, and v_1 has an empty body. v_1 stands for the parent while the parent runs f_1 inline, so that any `fork2join` nested inside f_1 uses the context and in-counter of v_1, and leaves those of v_0 alone. The steps are as follows.

(1) Save the parent's context in v_0`.k`. If v_2 is stolen, this is where the parent resumes.
(2) Set the in-counter of v_0 to 2, and schedule v_2, then v_1, which leaves v_2 stealable and v_1 not.
(3) Yield to the scheduler, saving the context in v_1`.k`. The scheduler pops v_1 and resumes it at once, so execution continues after the `swap`, with v_1 as the current vertex. Run f_1 inline.

```
type action = Init | Resume | Finish
type vertex = {in : atomic int, out : vertex*,
  act : action, k : context, body : () → void}
type worker = {k : context, cur : vertex*, ...}

loop() → void {   // on the worker's own stack
  while (!done) {
    v = pop()
    if (v == null) {
      v = steal()
      if (v != null) schedule(v)
      continue }
    me.cur = v
    if (v.act == Init) {
      v.k = new_context(new_stack(), enter)
      v.act = Finish }
    swap(me.k, v.k)   // run v until it yields
    if (v.act == Finish) notify(v) } }

enter() → void {   // first frame on a fresh stack
  me.cur.body(); restore(me.k) }

notify(v : vertex*) → void {
  u = v.out   // read before u.in can reach 0
  if (u == null) { done = true; return }   // root
  if (fetch_and_add(u.in, -1) == 1) schedule(u) }
```

Figure 1. The scheduler

```
fork2join(f1, f2 : () → void) → void {
  v0 = me.cur
  v1 = vertex{out = v0, body = λ () ⇒ {}}
  v2 = vertex{out = v0, body = f2}
  if (save(v0.k))                                    // (1)
    { v0.act = Finish; return }
  v0.act = Resume; v0.in = 2                         // (2)
  schedule(v2); schedule(v1)
  v1.act = Resume                                    // (3)
  swap(v1.k, me.k); v1.act = Finish
  f1()
  if (pop() == null) restore(me.k)                   // (4)
  v2.act = Resume; schedule(v2)                      // (5)
  swap(v2.k, me.k); v2.act = Finish
  f2()
  swap(v0.k, me.k); v0.act = Finish }                // (6)
```

Figure 2. Fork join

on task stack S	*in the scheduler of worker A*
(1) save(v_0.k)	
(2) schedule(v_2)	
schedule(v_1)	
(3) swap(v_1.k, A.k)	→ pop v_1, resume it
f_1()	
(4) pop() returns v_2	
(5) schedule(v_2)	
swap(v_2.k, A.k)	→ notify: v_0.in = 1
	pop v_2, resume it
f_2()	
(6) swap(v_0.k, A.k)	→ notify: v_0.in = 0
	schedule, pop, resume v_0
return	

worker A	*worker B*
(1)-(3); f_1() on S	steal v_2 from A
	fresh stack T; f_2() on T
(4) pop() returns **null**	
restore(A.k)	
notify: v_0.in = 1	
find other work ...	
	f_2() returns
	restore(B.k)
	notify: v_0.in = 0
	schedule, pop v_0
	restore(v_0.k): the save
	at (1) returns, on S

Figure 3. The two cases: v_2 not stolen (top), and v_2 stolen by worker B, which finishes last (bottom)

(4) Pop. If the pop fails, v_2 was stolen. Then abandon the current stack by switching to the scheduler without saving anything. The stack stays suspended at step 1, and the scheduler notifies v_0 that v_1 has finished.
(5) Otherwise, the pop returned v_2. Schedule v_2 again, so that it is the vertex scheduled most recently and cannot be stolen, and yield. The scheduler notifies v_0 that v_1 has finished, then pops v_2 and resumes it here. Run f_2 inline.
(6) Yield once more, saving the context in v_0.k. The scheduler notifies v_0 that v_2 has finished, which makes v_0 ready, then pops v_0 and resumes it here, and `fork2join` returns.

3 The two cases

Figure 3 traces both cases of `fork2join`.

Not stolen. One worker, A, does everything. f_1 and f_2 both run on the parent's task stack, S, as ordinary calls, and the only overhead is three round trips through the scheduler.

Stolen. Worker B steals v_2 from A while A runs f_1, gives v_2 a fresh stack, T, and runs f_2 there. When the pop at step 4 fails, A abandons S, and its scheduler notifies v_0 that v_1 has finished, which brings the in-counter of v_0 to 1. A then looks for other work. When f_2 returns, B's scheduler notifies v_0 that v_2 has finished, which brings the in-counter to 0. So B schedules v_0, pops it, and restores v_0.k. The call to `save` at step 1 returns a second time, on stack S as before, but now on worker B, and `fork2join` returns there. Had A finished last, A would have resumed the parent itself.

So a call to `fork2join` can begin on one thread and return on another. The stack stays the same throughout; only the thread that runs it changes.

4 Why it is safe

The correctness of the algorithm rests on four facts.

Only vertices that have never run are stolen. `fork2join` schedules v_1 (step 2) and v_2 (step 5) before it saves their contexts. At that moment, each is the vertex scheduled most recently, so it cannot be stolen, and its own worker pops it only after the `swap` to the scheduler has saved its context. The only vertices that thieves ever see are those like v_2 at step 2, whose action is `Init`.

Completion is reported from the scheduler. As soon as the in-counter of v_0 reaches zero, another worker may resume v_0 on stack S. So the decrement on behalf of v_1 must happen after A has left S, which is why step 4 switches to the scheduler instead of decrementing in place. Similarly, v_1 and v_2 live in the frame of v_0 on S, so `notify` reads the successor of a vertex before decrementing, and does not touch the vertex afterward.

The pop at step 4 is correct even if f_1 ***moved.*** f_1 returns on a different worker only if a vertex forked inside f_1 was stolen. Thieves take the oldest ready vertex, and v_2 is older than any vertex forked by f_1. So by then v_2 has been stolen as well, and the pop correctly fails.

A fresh stack is freed only after its last use. A stolen v_2 owns its fresh stack, and v_2 lives in the frame of v_0, so the stack is freed when the parent's call to `fork2join` returns. By then, f_2 has finished, and its worker has switched back to its scheduler.

5 Costs and consequences

Costs. A fork whose second branch is not stolen makes three round trips through the scheduler (steps 3, 5, and 6), six context switches in all, and allocates nothing, because v_1 and v_2 live in the frame of `fork2join`. In this case, no other worker can reach the in-counter of v_0, so TaskPaRTS updates it with plain loads and stores, rather than with atomic operations. The round trips are not essential to the technique. They keep the scheduler loop the only place where vertices start, resume, and finish, and they could be optimized away. As it is, how the six context switches are made matters more than any other choice on this path that we measured, and Appendix D records this choice and the others. A steal allocates one fresh

stack, 512 KB in TaskPaRTS, which is touched only as deep as f_2 recurses, and is freed when the parent returns from `fork2join`. Appendix E bounds the number of stacks in use.

Consequences for client code. Code that follows a call to `fork2join` may run on a different thread from the code that precedes it. So client code must not hold a lock across `fork2join`, nor expect a thread-local variable or the thread id to be the same afterward. More subtly, compilers assume that a function never changes threads, and so may reuse a thread-local address computed before the call. Then even a fresh read of a thread-local variable after `fork2join`, in the same function, can return the old thread's copy. Reading thread-local state through a function that the compiler can neither inline nor treat as pure avoids this problem. Other consequences follow from the use of task stacks. Task stacks have a fixed size, and no guard page by default. Exceptions must not escape f_1 or f_2, because unwinding cannot cross from a task stack into the scheduler. Backtraces end at the bottom of a task stack. And sanitizers, such as AddressSanitizer and ThreadSanitizer, are not told about the stack switches, so they may report false errors.

Implementations. TaskPaRTS implements the algorithm of Figures 1 and 2, and adds elastic scheduling, instrumentation, and CPU pinning around it. Micro taskparts implements the same algorithm, with a lock-free deque [3] that TaskPaRTS also uses, and nothing else, in about 600 lines of C++, including the assembly for x86-64 and AArch64. Appendix C maps the pseudocode to both.

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```
save:                       restore:
  movq %rbx, 0(%rdi)          movq 0(%rdi), %rbx
  movq %rbp, 8(%rdi)          movq 8(%rdi), %rbp
  movq %r12, 16(%rdi)         movq 16(%rdi), %r12
  movq %r13, 24(%rdi)         movq 24(%rdi), %r13
  movq %r14, 32(%rdi)         movq 32(%rdi), %r14
  movq %r15, 40(%rdi)         movq 40(%rdi), %r15
  leaq 8(%rsp), %rdx          movq 48(%rdi), %rsp
  movq %rdx, 48(%rdi)         movl $1, %eax
  movq (%rsp), %rdx           jmpq *56(%rdi)
  movq %rdx, 56(%rdi)
  xorl %eax, %eax
  ret
```

Figure 4. Saving and restoring a context on x86-64

```
type worker = {k : context, cur : vertex*,
  next : vertex*, dq : deque of vertex*}

schedule(v : vertex*) → void {
  if (me.next != null) push_bottom(me.dq, me.next)
  me.next = v }

pop() → vertex* {
  v = me.next
  if (v != null) { me.next = null; return v }
  return pop_bottom(me.dq) }

steal() → vertex* {
  return pop_top(random_other_worker().dq) }
```

Figure 5. Ready vertices: a private slot over a deque

A Machine contexts

A context holds the registers that the calling convention requires a callee to preserve [2, 11], plus the stack pointer and the address at which to resume:

	x86-64	AArch64
callee-saved	`rbx`, `rbp`, `r12`–`r15`	`x19`–`x29`, `d8`–`d15`
stack pointer	`rsp`	`sp`
resume address	return address	`lr` (`x30`)
total	8 words	21 words

Caller-saved registers need no saving: to the compiler, `save` is an ordinary call, so the compiler has already spilled any live caller-saved registers before calling it. Figure 4 shows both functions for x86-64. `save` records, as the stack pointer to restore, the value the stack pointer will have once `save` has returned, and, as the resume address, its own return address. `restore` reloads the registers, switches stacks, and jumps to the resume address with 1 in the return register. `swap`(k_1, k_2) is a single routine: the code of `save`, up to its return, followed by the code of `restore`, which reads k_2 from the second argument register, `%rsi`. The AArch64 versions are analogous. They too resume with an indirect branch to the restored `lr`, `br x30`, rather than with `ret` (Appendix D).

Declaring save to the compiler. Because `save` can return twice, it must be declared as such to the compiler (in GCC and Clang, with the attribute `returns_twice`). Otherwise, the compiler may keep a value that changes between the two returns in a callee-saved register, or reuse its stack slot for another value, and the second return then sees a stale or clobbered value. `swap` returns only once, when its context is restored, so it needs no such declaration.

Fresh contexts. To start a function f on a fresh stack s, the scheduler makes a context with `new_context`(s, f). The stack pointer of this context is the top of s, aligned to 16 bytes, its resume address is f, and its other registers are zero. On x86-64, the stack pointer is a further 8 bytes down, as if a `call` had pushed a return address. Restoring the context then starts f on s, as if f had been called there. TaskPaRTS builds fresh contexts differently, with the same effect: the scheduler loop calls `save`, and then overwrites the saved stack pointer with the top of a fresh stack. Restoring that context returns from that call to `save` on the fresh stack, where the code that follows calls `enter`.

B Ready vertices

Each worker keeps its ready vertices in a private slot, `next`, in front of a work-stealing deque, `dq` (Figure 5). `schedule`(v) puts v in the slot, and moves the slot's previous occupant to the bottom of the deque. `pop` takes from the slot first, and then from the bottom of the deque, and thieves take from the top of the deque. So the vertex scheduled most recently is always in the slot, where no thief can reach it, which is the property that Section 4 relies on.

At step 2 of `fork2join`, the slot is empty: the current vertex was itself popped from it, and any nested `fork2join` calls have taken back what they scheduled. So `schedule`(v_2) puts v_2 in the slot, and `schedule`(v_1) moves v_2 to the deque. The slot is also empty whenever `notify` or a thief schedules a vertex, so a vertex that reaches the deque is always a v_2 displaced at step 2.

Any work-stealing deque will do. Micro taskparts uses the lock-free deque of Arora et al. [3], which TaskPaRTS also uses when elastic scheduling is disabled, unless it is configured to use the deque of Chase and Lev [6]. With elastic scheduling, TaskPaRTS uses a deque by Yao, Westrick, Rainey, and Acar, which supports it.

C From pseudocode to code

Table 1 maps the names in the pseudocode to their counterparts in micro taskparts and in TaskPaRTS. Both are in the TaskPaRTS repository [12]: micro taskparts in the directory

pseudocode	micro taskparts	TaskPaRTS
save, restore	mtp_context_save, mtp_context_restore	context_save, context_restore
swap	mtp_context_swap	swap
vertex	vertex	native_fork_join_vertex
action	action	continuation_action
next, dq	next, deque	bottom, deque
deque	abp	abp, chaselev, ywra
loop	worker_loop	loop
new_context	new_context	new_continuation
enter	vertex_entry	enter
notify	notify	parallel_notify
fork2join	fork2join	fork2join

Table 1. Names in the pseudocode, in micro taskparts, and in TaskPaRTS

doc/native-fork-join/microtaskparts, and TaskPaRTS in the files src/taskparts.cpp and include/taskparts/taskparts_internal.hpp. Both implementations number the steps of fork2join as in Figure 2.

Beyond micro taskparts, TaskPaRTS adds the following.

- A choice of deques, including one that supports elastic scheduling (Appendix B).
- Elastic scheduling: workers that fail to find work suspend on a semaphore, instead of spinning, and are woken when there is surplus work.
- Non-atomic in-counter updates when the second branch is not stolen (clone_fast).
- A persistent pool of workers, and a sink vertex that marks the end of each top-level computation, in place of the done flag.
- Statistics, logging in a format that the Perfetto trace viewer reads, and CPU pinning with hwloc.

D Performance notes

A fork whose second branch is not stolen does little work, so small choices in the implementation show up in the running time. This appendix records the choices that we measured with micro taskparts, running fib(38), which makes 63 million forks, on an Apple M5 with 4 performance and 6 efficiency cores (Table 2). Each time in the table is the median of five runs, and the runs of the six versions were interleaved. Flexibench recorded the runs, checked that every version computes correct results, and certified the values in the table [13]. The record is in the directory doc/native-fork-join/perf of the TaskPaRTS repository [12]. TaskPaRTS makes the same choices, except where noted.

A lock-free deque. With a deque protected by a lock, a fork takes two uncontended locks, at steps 2 and 4. For fib, this was not the bottleneck: the lock-free deque of Appendix B is about 10% faster on one worker. But the lock had hidden the next problem.

	workers		
version	1	4	8
deque protected by a lock	5.60	1.47	1.24
lock-free deque	4.98	2.72	3.16
+ victims chosen by hashing	4.94	1.31	0.86
+ one cache line per vertex field	4.93	1.30	0.88
+ fewer lookups of the worker	4.54	1.17	0.74
+ context switch in one call	1.85	0.49	0.33

Table 2. Running time of fib(38), in seconds, with micro taskparts, as each change is applied in turn: the median of five runs

A slowdown with several workers. In the first two versions, each worker chose its victims with a generator of its own, of type std::minstd_rand. With the lock-free deque, fib(38) then took 2.72 s on 4 workers, about twice as long as with the next version (1.31 s), which differs only in choosing victims by hashing the worker's id and a count of its steal attempts, as TaskPaRTS does. On 8 workers, it took longer still (3.16 s). Yet workers attempted only a few hundred steals per run, and, in the common case, no two workers touch the same vertex. The slowdown persisted even when we allowed a single steal, after which the two workers ran disjoint subcomputations. In earlier measurements, outside the table, other changes also removed the slowdown while keeping the generator: starting the in-counter, the out-set, and the context of a vertex each on a cache line of its own, as in TaskPaRTS; updating the in-counter on the fast path with plain loads and stores, rather than with an atomic decrement, which TaskPaRTS also does (clone_fast); and adding any atomic operation just before the pop at step 4, even one with no memory ordering, which is how the lock had hidden the slowdown. These changes have little in common, and we have not found the cause. The slowdown seems to depend on details of the code and of the layout of data, rather than on the algorithm, and so it may reappear after unrelated changes. Once victims are chosen by hashing, one cache line

per vertex field makes no measurable difference, but micro taskparts keeps it, as TaskPaRTS does.

Fewer lookups of the worker. Each use of `me` looks up the worker in thread-local storage, through a function that the compiler can neither inline nor treat as pure (Section 5). The compiler barrier that enforces this emits no instruction. The cost is the call and the lookup, which on macOS is itself a call. But in `fork2join`, the worker changes only across the calls to f_1 and f_2, and across the second return from `save` at step 1. The switches to the scheduler at steps 3, 5, and 6 resume the vertex on the same worker, because the vertex is in that worker's private slot. And the scheduler loop never changes worker. Looking up the worker once per stretch, rather than at each use, cuts the lookups per fork from about 12 to 3, and the running time on one worker by about 8%. TaskPaRTS looks up the worker at each use.

A context switch in one call. In the other versions, a context switch is two calls, `save` and then `restore`, as in the definition of `swap` in Section 2, and, on AArch64, `restore` resumes with `ret`. The processor predicts where a `ret` goes from a stack of the return addresses of recent calls, and after a context switch, this prediction is wrong. The last version switches with a single routine (Appendix A), and resumes with an indirect branch, `br x30`, instead. This makes `fib(38)` more than twice as fast on 1, 4, and 8 workers. On one worker, it takes 1.85 s instead of 4.54 s, or about 29 ns per fork instead of 72. In measurements outside the table, each half of the change helped on its own, the single routine more than the indirect branch, but we have not established why the single routine helps as much as it does. TaskPaRTS still switches with two calls, and on AArch64 resumes with `ret`. On x86-64, both already resume with an indirect jump (Figure 4), and we have not measured the single routine there.

E Native fork join and the cactus-stack problem

This appendix places native fork join among other ways of managing stacks under work stealing, starting from the survey of Lee et al. [10].

The cactus-stack problem. Lee et al. ask for three properties at once. The first is *serial-parallel reciprocity*: parallel code and serial code, including legacy binaries compiled for an ordinary linear stack, can call each other freely. The second is a time bound $T_P \le T_1/P + c_\infty T_\infty$, where T_1 is the work, T_∞ the span, and P the number of workers. The third is a space bound $S_P \le PS_1$, where S_1 is the stack space of a serial execution. Meeting all three is what they call the *cactus-stack problem*. They survey seven strategies, which Table 3 summarizes.

Strategy 1, used by Cilk-5, allocates every frame on the heap, which requires recompiling all code, serial code included. Strategies 2 to 6 keep linear stacks. They differ in what a worker does when it must steal while its stack holds a frame that it cannot pop, because a thief is running the continuation of that frame. The worker may steal onto the same stack anyway (2), steal only frames deeper than those on its stack (3), stop stealing while its stack is too deep (4), take a new stack (5), or take the stack of an ancestor of the stolen frame, if that ancestor is suspended with no frame above it (6). Strategy 7, Cilk-M, maps the frames that a thief shares with its victim into the stack of each, at the same virtual addresses, which requires support from the operating system.

strategy	recip.	time	space
1. Recompile everything	no	very strong	very strong
2. One stack per worker	yes	very strong	no
3. Depth-restricted stealing	yes	no	very strong
4. Limited-depth stacks	yes	no	very strong
5. New stack when needed	yes	very strong	weak
6. Recycle ancestor stacks	yes	strong	weak
7. TLMM cactus stacks	yes	strong	strong
Native fork join	yes	very strong[†]	weak

Table 3. Strategies for managing stacks under work stealing, from Lee et al. [10], with native fork join added: whether each provides serial-parallel reciprocity, and how strong its time and space bounds are. [†]Expected, but not proved.

Where native fork join fits. Native fork join is not orthogonal to these strategies: it manages stacks in the same way as Strategy 5. A thief always runs a stolen vertex on a fresh stack. A worker that reaches a join whose second branch is still running leaves its stack suspended and returns to its scheduler, so whatever it steals next runs on another fresh stack. And a stack is freed only once it holds no frame that can still run. What is new is how native fork join gets there: by stealing children rather than continuations, and by switching contexts, which needs no support from the compiler or the operating system.

Children, not continuations. Lee et al. assume continuation stealing, as in Cilk, and set child stealing aside, because it can need unbounded bookkeeping, even on one processor. That happens when a parent can keep forking before it runs any child, as in a loop of spawns. Binary fork join rules this out: a fork pushes one vertex and runs f_1 at once, so a worker's deque holds at most one vertex per call to `fork2join` in progress on its stack, just as a Cilk deque holds at most one frame per spawn in progress. In fact, for binary fork join, the two disciplines steal the same work. `fork2join(`f_1`, `f_2`)` corresponds to spawning f_1, calling f_2, and syncing, and the stolen v_2 corresponds to the continuation of the spawn, up to the sync. In both, whoever reaches the join last runs the code that follows it, which is why that code can run on a different thread from the code that precedes the fork.

No frame spans two stacks. The difference that matters is how a computation moves from one worker to another. Under continuation stealing, a thief resumes the stolen frame in place. In Strategy 2, for example, the thief points its base pointer at the frame, which stays on the victim's stack, and its stack pointer at its own stack, where the frame's callees then run. So parallel code must be compiled to work with a frame and its callees on different stacks, or, in Cilk-M, separated by a gap. Lee et al. list seven invariants that compiled Cilk-M functions must maintain, and hand-compiled their benchmarks to meet them.

Native fork join never splits a frame between stacks. A thief runs only the thunk $\texttt{f}_2$, as an ordinary call on its own stack, and the parent resumes on its own stack, by restoring a context saved by an ordinary call to `save`. The only pointers from one stack into another are those that client code creates, such as the references that $\texttt{f}_2$ captures, which are ordinary C++. So native fork join provides Strategy 5 in a library, with an unmodified compiler and operating system. Serial-parallel reciprocity holds in the sense of Lee et al.: `fork2join` is an ordinary function, which any serial code can call, including a legacy binary that calls back into client code. Two caveats from Section 5 fall outside their criteria. The code after a join may run on another thread, which is also true under continuation stealing. And the frames on a task stack must fit in its fixed size.

Time. Under native fork join, no worker ever waits at a join, and a steal costs one stack allocation. Moreover, execution fits the DAG model analyzed by Arora et al. [3]. A vertex that forks pushes one of its successors and runs the other next, and a vertex that a join makes ready runs next on the worker that made it ready, through the private slot of Appendix B. So we expect native fork join to meet the time bound of Strategy 5, with a constant c_∞, but we have not written out a proof. By contrast, library schedulers that help while waiting, such as those of TBB and ParlayLib (Section 1), keep one stack per worker, as Strategies 2 to 4 do. Lee et al. liken the heuristic of TBB to Strategy 3, which can lose the time bound.

Space. Native fork join pays the price of Strategy 5. As Lee et al. note for Strategy 5, the busy-leaves property holds: a stack suspended at a join waits on a stolen vertex, whose stack is either running or itself suspended at a join, and so on, down to a stack that a worker is running. So the stacks in use form a tree with at most P leaves. Each stack starts at a vertex nested more deeply in calls to `fork2join` than the vertex that starts its parent, so the tree has depth at most D, the maximum nesting depth of calls to `fork2join`, and there are $O(PD)$ stacks in use. This count holds even if the stack of a stolen vertex counts as in use until its parent returns from `fork2join` (Section 4). Lee et al. bound the physical memory that Strategy 5 uses for frames by PS_1. But a stack can grow as large as S_1, and is recycled only when it holds no frames, so stacks can take up to DPS_1 of virtual memory. In native fork join, each stack has a fixed size Z, 512 KB in TaskPaRTS, so stacks take $O(PDZ)$ of virtual memory. And a page of a stack, once touched, stays in memory at least until the stack is freed.

Refinements. Lee et al. mention two refinements of Strategy 5 that appear to carry over to native fork join, and that neither TaskPaRTS nor micro taskparts implements. One is to unmap the pages of frames once they are popped, to give their physical memory back, which they expect to be costly unless done lazily. The other is Strategy 6, in which a thief runs a stolen vertex on the stack of a suspended ancestor, when no frame lies above that ancestor's frame. This is safe, because the ancestor cannot resume before the stolen vertex finishes. But Lee et al. report that Strategy 6 can still need $\Omega(P^2S_1)$ stack space.

Stacks per steal, not per thread. A library of user-level threads typically gives each thread a stack when the thread is created. Native fork join creates a stack only when a vertex is stolen; a vertex that is not stolen runs on the stack of its parent. So a computation allocates one stack per steal, rather than one per fork, and work stealing makes $O(PT_\infty)$ steal attempts in expectation [3, 5].